

CivicCompassHANNegotiator for ANAC 2026 HAN

Abstract

CivicCompassHANNegotiator is a hybrid negotiator for the ANAC 2026 Human-Agent Negotiation League. It combines a utility-maximizing alternating offers policy with human-readable Markdown messages. The central design goal is to perform well against LLM-simulated and crowd-sourced human partners while remaining reliable in the tournament environment. Formal offers and accept/reject decisions are deterministic; an optional local LLM is used only to refine message wording. Version 0.0.4 also exports a compatible human-preference estimate and reacts to credible late final-offer language.

Setting

HAN uses the alternating offers protocol augmented with text communication. At each turn, the agent must decide whether to accept the current proposal, reject with a counteroffer, or end the negotiation. In addition to utility, human partners evaluate the negotiation experience, so the agent must provide clear and respectful explanations rather than only numerical proposals.

Opponent Modeling

The agent enumerates the finite outcome space and keeps all outcomes above its reservation value. It first estimates its own issue sensitivity by marginalizing utility over each issue value. This supports a complement prior: values that are cheap for the agent are initially treated as likely valuable to the human. This prior is well matched to distributive allocation and buyer-seller trade settings, while still being corrected online by observed behavior.

Human offers are then used to build an online ordinal model of the partner’s likely preferences. Encoded numeric outcomes such as 151_148 are expanded into numeric features. For numeric issues, the agent applies proximity smoothing

$$K(a, b) = \exp(-|a - b|/\sigma),$$

so that nearby values share evidence. Early offers are weighted more strongly than late concessions because they are more likely to reveal the partner’s preferred region. The resulting model is published as a `MappingUtilityFunction` through common opponent-model compatibility methods and private-info keys.

Text-Based Preference Signals

HAN partners can communicate in free text. The agent parses incoming text for issue names, issue values, priority words such as “need”, “important”, “fair”, and “balanced”, numeric-direction cues such as “too high” or “more”, and finality cues such as “final offer” or “best I can do”. Mentioned issues receive additional attention in the human model, and the value attached to the partner’s current formal offer is boosted or shifted in the indicated direction. Finality cues only affect late acceptance, preventing avoidable no-deal outcomes without weakening early bargaining.

Bidding Policy

The bidding policy uses an aspiration curve

$$A(t) = r + (u^{\max} - r)(1 - 0.055t - 0.905t^{3.85}),$$

where r is the reservation value. Candidate bids are selected from outcomes above or near this aspiration level and filtered through a Pareto-style frontier. They are then ranked by own normalized utility, the modeled human preference score, Nash-style joint surplus, a deterministic face score, and a repeat penalty. This produces offers that retain utility while remaining interpretable and plausible to a human partner.

Text Generation

Messages are deterministic templates grounded in the actual counteroffer. The agent explains one or two changed issues, distinguishes opening, middle, and late negotiation phases, and avoids unsupported claims. When possible, it names a trade: where it is giving ground and where it still needs movement. The text is concise, cooperative, and framed around reaching a workable agreement.

If local Ollama is available, the agent calls `qwen3:4b-instruct` with a short timeout to refine only the wording of the outgoing message. The prompt forbids changing the formal offer, revealing numeric utilities, or producing internal reasoning. If the call fails or the output violates these constraints, the deterministic message is used unchanged.

Acceptance

The acceptance policy compares each incoming offer to the planned counteroffer and the current aspiration level. It also enforces a separate deal-quality floor that decays slowly over time. Near the deadline, the agent therefore relaxes only for offers that still preserve meaningful surplus above reservation, reducing unnecessary breakdowns without buying agreement at very low utility. Credible late final-offer language slightly relaxes this floor above reservation, which is useful against human or LLM partners that stop moving before the formal deadline.

Expected Competitive Properties

- Against concessionary humans, the agent starts with high-utility offers but accepts strong proposals promptly.
- Against stubborn humans, the agent explains concrete issue changes and concedes in a controlled way.
- Against LLM-simulated humans, the messages provide clear tradeoff rationales and can use local LLM wording refinement without giving the model control over decisions.
- Against varied scenarios, numeric feature expansion improves generalization across resource distribution and trade-like domains.

Validation

The implementation follows the official HAN template API by returning `SAOResponse` objects with optional Markdown text in `data["text"]`. It imports successfully under the installed NegMAS environment, has no hard dependency on LLM packages, and is packaged with a two-page report and pinned requirements for submission. Smoke tests covered the public Grocery, Island, and Trade scenarios as well as generated NegMAS domains.